Answer key (Module 15 exercises)

```spl
index=_internal sourcetype=splunkd_access earliest=-24h
| stats count by user
| lookup users_lookup username AS user OUTPUT department, criticality
| where criticality="high"
```

```spl
index=_internal sourcetype=splunkd log_level=ERROR
[ search index=_internal sourcetype=splunkd log_level=ERROR | top limit=1 host | fields ho
| head 20
```

```spl
index=_internal sourcetype=splunkd
| eval is_err=if(log_level=="ERROR",1,0)
| eventstats sum(is_err) AS host_errors by host
| eventstats max(host_errors) AS max_errors
| where host_errors=max_errors
| head 20
```

```spl
index=_internal sourcetype=splunkd_access
| sort 0 _time
| streamstats time_window=2m count AS reqs by clientip
| where reqs>10
```

```spl
| tstats latest(_time) AS last_seen WHERE index=* BY host
| eval mins_silent=round((now()-last_seen)/60)
| where mins_silent>30
```

📘 MODULE 16: DATA MODELS

16.1 Concept

A **data model** is a structured, hierarchical description of your data. It defines *datasets* (a root dataset plus children), the fields each one has, and the search constraint that selects the right events. It doesn't store data. It is a saved **view** of your events.

```
Authentication                  ← root dataset (constraint: tag=authentication)
 ├── Failed_Authentication      ← child (adds: action=failure)
 └── Successful_Authentication  ← child (adds: action=success)
```

16.2 Why data models exist

1. **Normalization:** all data that is "authentication" (Linux, Windows, VPN) looks the same to a search.
2. **Speed:** with **acceleration**, `tstats` can query them very fast.
3. **Reuse:** dashboards, ES correlation searches, and Pivot read from them, so analysts don't write raw SPL for every source.

16.3 Parts of a data model

Part	Meaning
Root dataset	Top-level dataset. Types: **event** (most common), **search**, **transaction**
Child dataset	Inherits the parent's constraints and fields and adds its own filter
Constraint	The base search that selects events, e.g. `index=_internal sourcetype=splunkd`
Fields	Auto-extracted, eval expressions, lookups, regex extractions, and others
Acceleration	Splunk pre-builds tsidx summaries of the data model on the indexers

16.4 Acceleration (the key to speed)

- When enabled, Splunk runs background searches that write **summary tsidx files** next to the index buckets.
- `tstats summariesonly=true` reads **only those summaries**. It is fast, but results lag until the summary is built.
- **Costs:** extra disk and CPU. Accelerate only the data models you use, with a limited summary range (e.g. 7 or 30 days).
- Enable it in **Settings → Data models → (model) → Edit → Edit Acceleration**. Only event datasets (and streaming-only search datasets) can be accelerated.

16.5 The connection (memorize)

```
DATA MODEL → ACCELERATION → TSTATS → FASTER SEARCH
```

16.6 Commands

Query 1: `datamodel` command

spl

```
| datamodel Authentication Authentication search
| head 10
```

What it does: runs the data model's root dataset like a search and returns events.
Line-by-line: `datamodel` is generating (leading `|`) · first name is the **data model**, second is the **dataset** · `search` returns events. (`| datamodel` alone lists all models.)
Common mistake: wrong dataset name, or the CIM app isn't installed.
Interview explanation: "I use `datamodel` to check what a model returns before building `tstats` searches on it."

Query 2: `tstats` on a data model

```spl
| tstats summariesonly=true count
  FROM datamodel=Authentication
  WHERE Authentication.action="failure"
  BY Authentication.src, Authentication.user
| rename Authentication.* AS *
| sort - count
```

Line-by-line: `FROM datamodel=` picks the model · fields use the **`Dataset.field`** **prefix** · `rename` removes the prefix for readability.
Expected result: a table of `src`, `user`, `count`.
Common mistakes: no prefix on fields, `summariesonly=true` before acceleration finishes, and the data isn't tagged for the model.

Query 3: child dataset

```spl
| tstats count FROM datamodel=Authentication.Failed_Authentication BY Authentication.src
```

Or with the root datamodel and a `nodename` filter. The exact child dataset names depend on your CIM version, so check them in Settings → Data models.

16.7 Hands-on lab (works without CIM)

1. **Settings → Data models → New Data Model.** Title: `Splunk_Internal`.
2. **Add Dataset → Root Event.** Name it `splunkd_events`, with constraint `index=_internal sourcetype=splunkd`.
3. **Add fields:** `log_level`, `component` (use "Auto-extracted" and pick them). Save.
4. Test:

```spl
| datamodel Splunk_Internal splunkd_events search | head 5
```

5. Query it with `tstats` (not accelerated, so use `summariesonly=false`):

```spl
```

```
| tstats summariesonly=false count FROM datamodel=Splunk_Internal
  WHERE splunkd_events.log_level="ERROR"
  BY splunkd_events.component
```

Use the dataset name exactly as the editor shows it.
6. Optional: enable acceleration and compare speed in the job inspector.

16.8 Troubleshooting

Symptom	Cause
`tstats` on a model returns nothing	Not accelerated + `summariesonly=true`, constraint matches no events, wrong field prefix, or no data tagged for the model
Recent events missing	Summary not caught up yet
Model returns data in Pivot but not `tstats`	Field not part of the model, or acceleration disabled
Acceleration using lots of disk	Summary range too long, or too many models accelerated

16.9 Interview Q&A

- **What is a data model?** A hierarchical, structured definition of datasets (constraints + fields) that sits on top of raw events.
- **Why accelerate it?** To build tsidx summaries so `tstats` is very fast.
- **What does `summariesonly=true` do?** It reads only accelerated summaries, so unsummarized data is not searched.
- **Root vs child dataset?** The child inherits the parent's constraints and adds its own filter.
- **Why the `Dataset.field` prefix?** `tstats` references fields by their data model path.
- **Does a data model store data?** No. It is a definition, and acceleration stores *summaries*.

16.10 Revision

Data model = datasets + constraints + fields · acceleration = summaries · `tstats` + `summariesonly` · prefix fields · `datamodel` command for testing.